ITN 7.02 – Case Study - Part 4 (Modules 11 – 13)

Your supervisor has asked you to prepare a fourth part for the network simulation and presentation you have been working on. You will continuing with building the network you created for Part 3. You will add on to this file for Part 4. Open the Packet Tracer file **ITN – Part 4**. Be sure to do a **Save As** to rename the new file. A presentation will be used to demonstrate to the potential customer how your company would design and implement the proposed network. You have been asked to design this project in several segments that can ultimately be assembled into the full network topology.

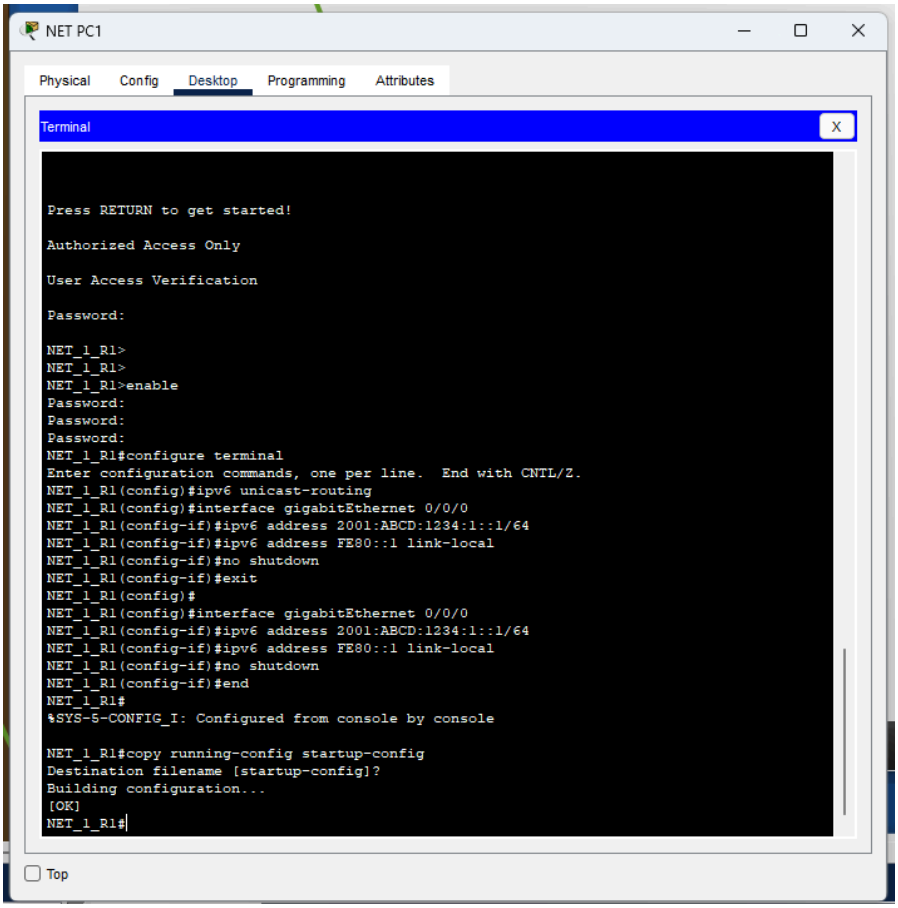
Part 4 (Modules 11 – 13)

Step 1 – Designing and documenting an IPv4 subnet scheme for a future growth network

- You have been asked to plan an IPv4 address scheme for a future network.
- You have been given the IPv4 address range **192.168.1.0 /24** to address the network.
- Use this address space to create an address table bases on the following requirements:
    - LAN-A will need **60** host addresses.
    - LAN-B will need **25** host addresses.
    - LAN-C will need **20** host addresses.
    - LAN-D will need **12** host addresses.
    - LAN-E will need **12** host addresses.
    - The network will also need to address four point-to-point links.
- Include this table in your presentation and be prepared to explain how you develop it.

Step 2 – Configuring IPv6 on the router

- Access the router **NET_1_R1** via the **NET PC1** terminal program.
- Enable IPv6 routing
- Configure interface G0/0/0 with an IPv6 address
    - GUA – 2001:ABCD:1234:1::1 /64
    - Link local – FE80::1
- Configure interface G0/0/1 with an IPv6 address
    - GUA – 2001:ABCD:1234:2::1 /64
    - Link local – FE80::1



ภาพแสดงการตั้งค่า IPv6 บนราวเตอร์ NET_1_R1 ผ่าน Terminal มีขั้นตอนดังนี้

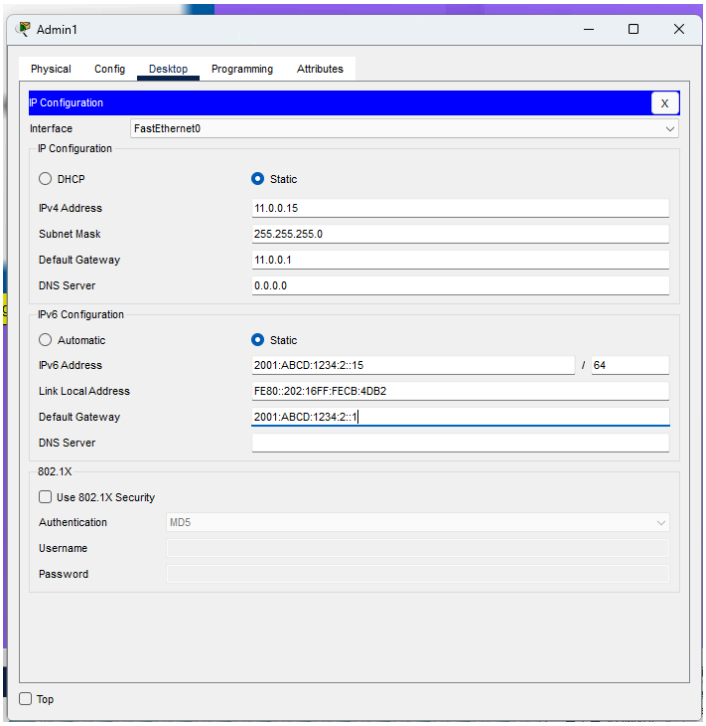
1. **Login** เข้าเราเตอร์ด้วยรหัสผ่าน แล้วพิมพ์ enable เพื่อเข้าโหมดสิทธิ์สูง
2. configure terminal เข้าสู่โหมดตั้งค่า
3. ipv6 unicast-routing เปิดใช้งาน IPv6 บนราวเตอร์
4. ตั้งค่าพอร์ต G0/0/0: ใส่ ipv6 address 2001:ABCD:1234:1::1/64 และ FE80::1 link-local แล้ว no shutdown
5. ตั้งค่าพอร์ตที่สอง (ควรเป็น G0/0/1 กับ address 2001:ABCD:1234:2::1/64 แต่ในภาพพิมพ์ค่าซ้ำกับพอร์ตแรก — จุดนี้น่าจะพิมพ์ผิด)
6. end ออกจากโหมดตั้งค่า
7. copy running-config startup-config บันทึกค่าลงหน่วยความจำถาวร ระบบขึ้น [OK] แสดงว่าสำเร็จ

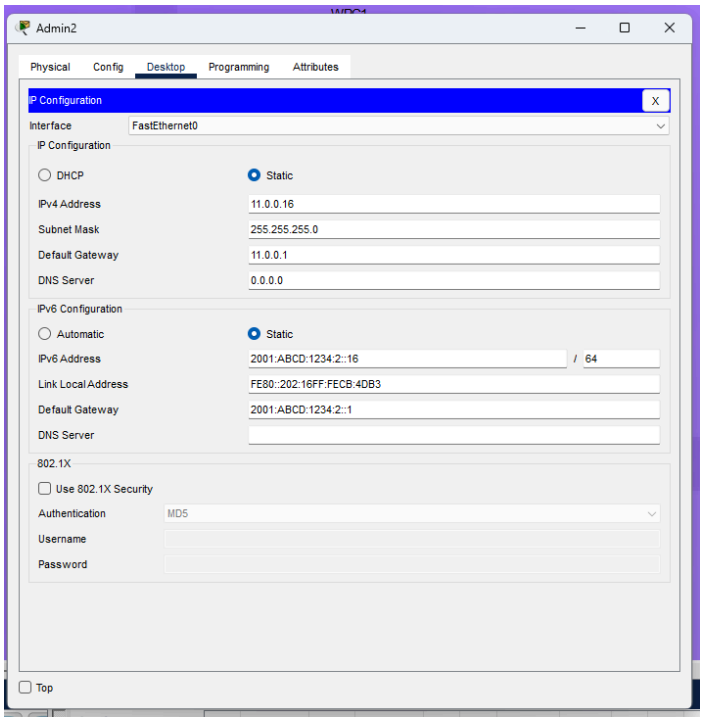
Step 3 –Configure PCs with an IPv6 address

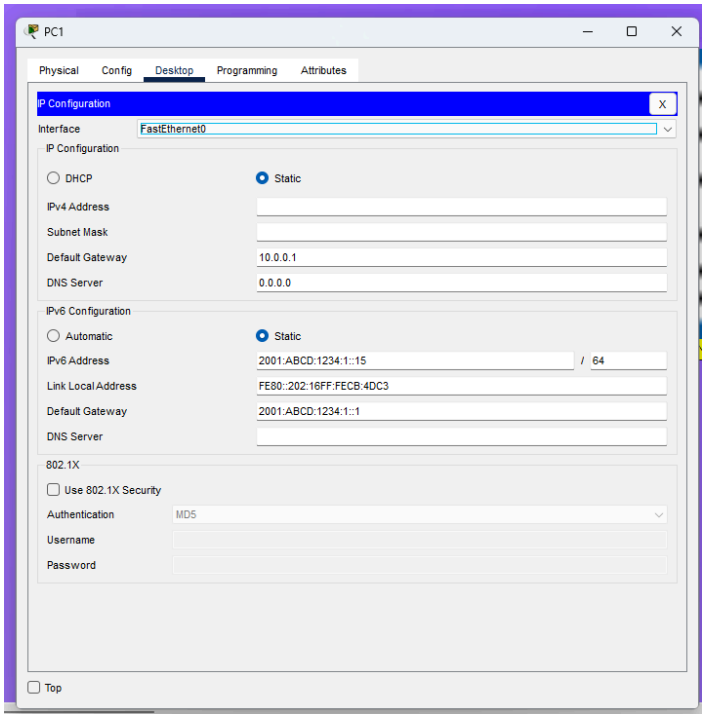
- PC1
  - GUA – 2001:ABCD:1234:1::15 /64
  - Link local – FE80::15

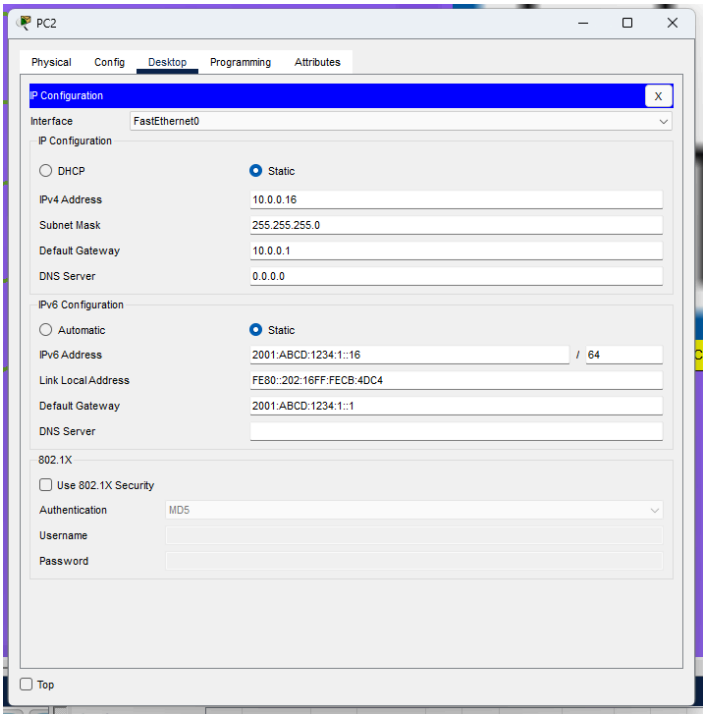
    - Use the address of the G0/0/0 interface as the default gateway
- PC2
    - GUA – 2001:ABCD:1234:1::16 /64
    - Link local – FE80::16
    - Use the address of the G0/0/0 interface as the default gateway
- PC3
    - GUA – 2001:ABCD:1234:1::17 /64
    - Link local – FE80::17
    - Use the address of the G0/0/0 interface as the default gateway
- PC4
    - GUA – 2001:ABCD:1234:1::18 /64
    - Link local – FE80::18
    - Use the address of the G0/0/0 interface as the default gateway
- PC5
    - GUA – 2001:ABCD:1234:1::19 /64
    - Link local – FE80::19
    - Use the address of the G0/0/0 interface as the default gateway
- Admin1
    - GUA – 2001:ABCD:1234:2::15 /64
    - Link local – FE80::15
    - Use the address of the G0/0/1 interface as the default gateway
- Admin2
    - GUA – 2001:ABCD:1234:2::16 /64
    - Link local – FE80::16
    - Use the address of the G0/0/1 interface as the default gateway

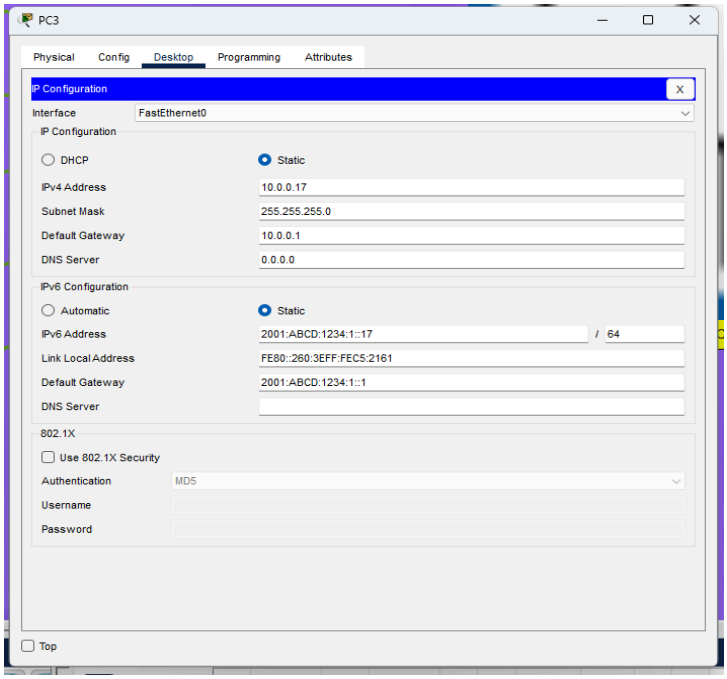
ภาพแสดงการIP Config ของ AdminและPC

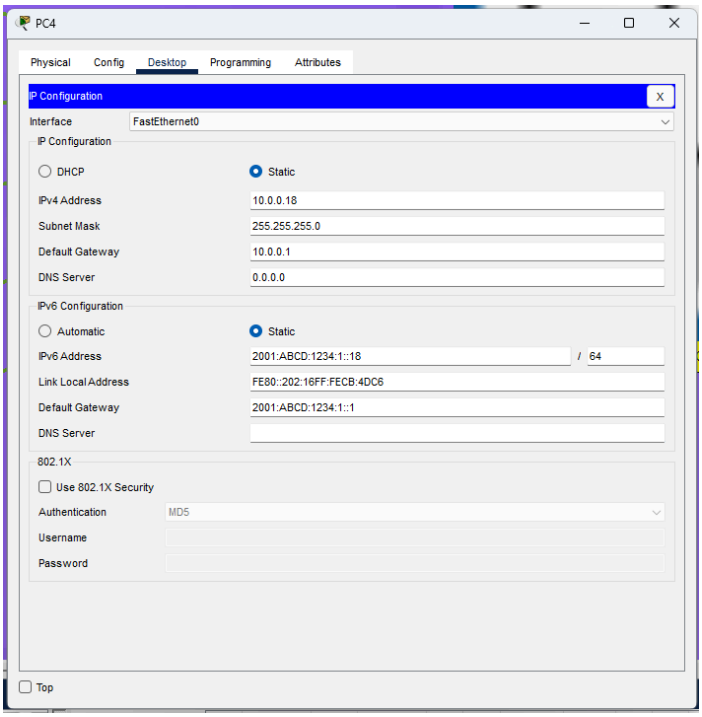

Admin1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
11.0.0.15
Subnet Mask
255.255.255.0
Default Gateway
11.0.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:ABCD:1234:2::15
/ 64
Link Local Address
FE80::202:16FF:FECB:4DB2
Default Gateway
2001:ABCD:1234:2::1
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

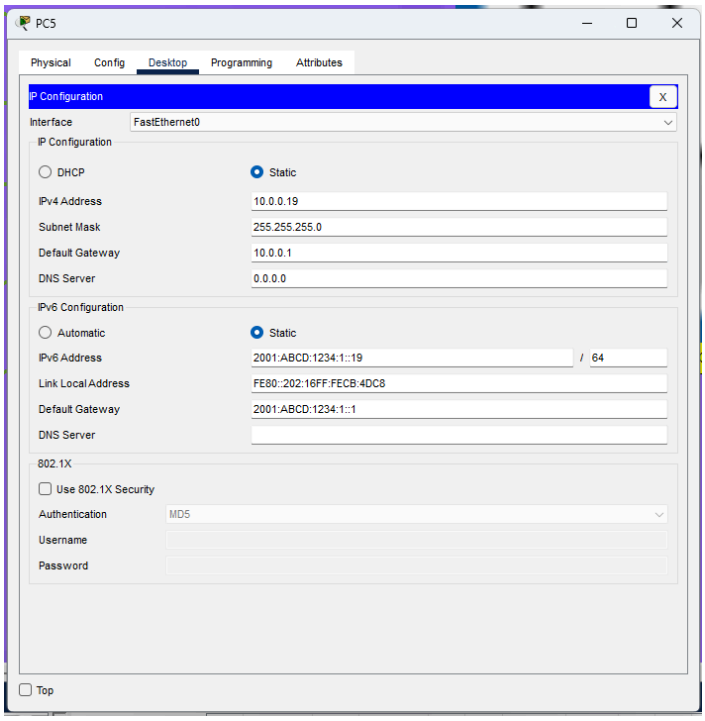

Admin2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
11.0.0.16
Subnet Mask
255.255.255.0
Default Gateway
11.0.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:ABCD:1234:2::16
/ 64
Link Local Address
FE80::202:16FF:FECB:4DB3
Default Gateway
2001:ABCD:1234:2::1
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
10.0.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:ABCD:1234:1::15
/ 64
Link Local Address
FE80::202:16FF:FECB:4DC3
Default Gateway
2001:ABCD:1234:1::1
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.0.0.16
Subnet Mask
255.255.255.0
Default Gateway
10.0.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:ABCD:1234:1::16
/ 64
Link Local Address
FE80::202:16FF:FECB:4DC4
Default Gateway
2001:ABCD:1234:1::1
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top



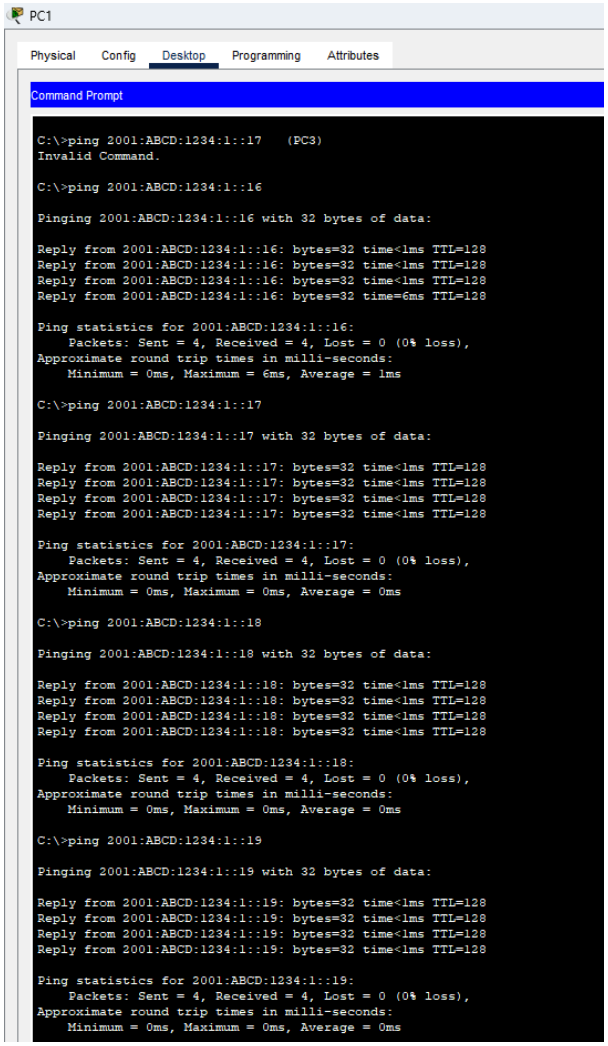


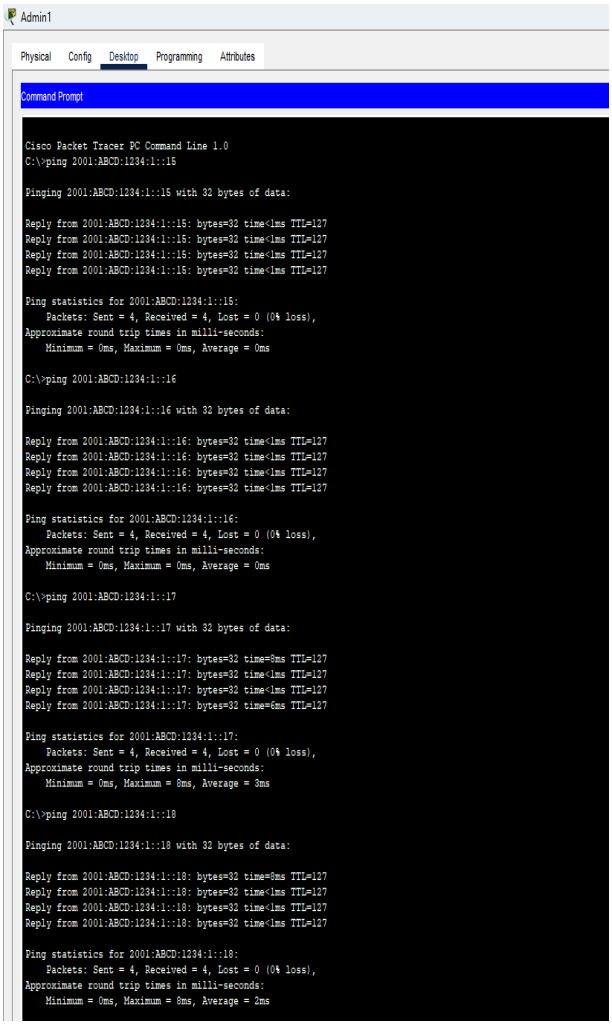


Step 4 – Verify IPv6 connectivity

- Use PING to verify host-to-host connectivity.
- From PC1 test IPv6 connectivity to all other PCs with IPv6 addresses.
- From PC2 test IPv6 connectivity to all other PCs with IPv6 addresses.
- From PC3 test IPv6 connectivity to all other PCs with IPv6 addresses.
- From PC4 test IPv6 connectivity to all other PCs with IPv6 addresses.
- From PC5 test IPv6 connectivity to all other PCs with IPv6 addresses.

- From Admin1 test IPv6 connectivity to all other PCs with IPv6 addresses.
- From Admin2 test IPv6 connectivity to all other PCs with IPv6 addresses.





ภาพแสดงผลการทดสอบ **Ping** เพื่อตรวจสอบการเชื่อมต่อ IPv6 จากเครื่อง PC1 และ Admin1

PC1 (ซ้าย)

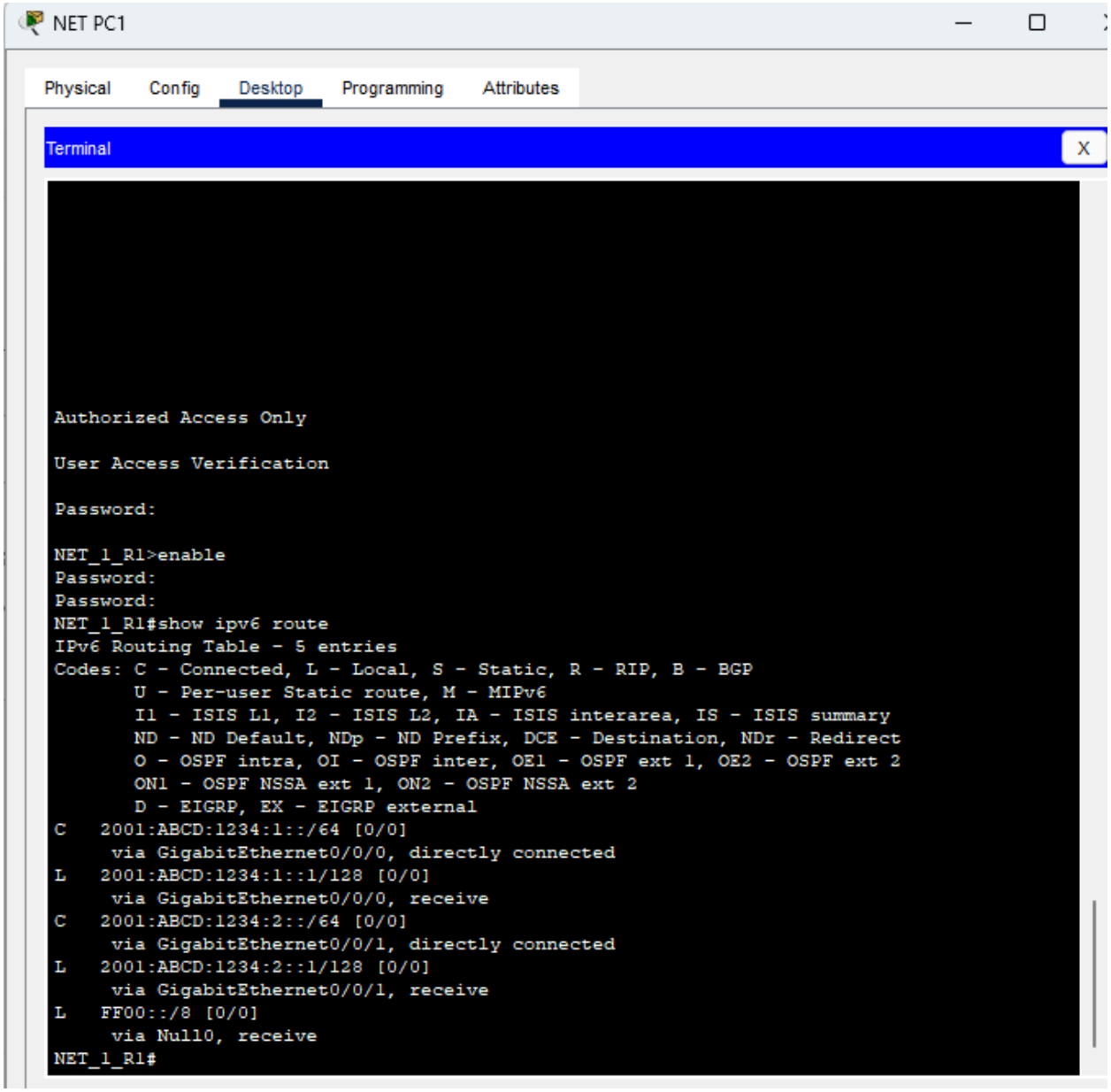
- Ping ไป 2001 : ABCD : 1234 : 1 : : 17 ครั้งแรกขึ้น "Invalid Command" (พิมพ์ผิดหรือมีวงเล็บ (PC3) ติดไปด้วย)
- Ping ไป : : 16 (PC2), : : 17 (PC3), : : 18 (PC4), : : 19 (PC5) สำเร็จทุกรายการ ได้ Reply ครบ 4/4 แพ็กเก็ต ไม่มี packet loss (0%)

Admin1 (ขวา)

- Ping ไป 2001 : ABCD : 1234 : 1 : : 15 (PC1), : : 16 (PC2), : : 17 (PC3), : : 18 (PC4) สำเร็จทุกรายการเช่นกัน ได้ Reply ครบ 4/4 แพ็กเก็ต ไม่มี packet loss (0%)

Step 5 – Verify IPv6 routing

- Verify the IPv6 routing table on **NET_1_R1**.



ภาพแสดงผลลัพธ์คำสั่ง `show ipv6 route` บนราวเตอร์ NET_1_R1 เพื่อตรวจสอบตาราง IPv6 Routing

Step 6 – Documentation

- Add to the documentation you previously created of the physical network. Remember you will be adding to this document throughout the case study, so consider how you want it to be laid out as the potential customer will likely want to review the document.
- Document the current IPv6 routing table.
- Document the configuration of **NET_1_R1** (text copy).
- Document all IP addressing on the PCs.

Step 7 – Additional information for presentation

- Explain the role of the subnet mask when assigning IPv4 addresses.
- Explain how to determine what network a host belongs to.
- Explain the differences in
    - Unicast transmissions
    - Multicast transmissions
    - Broadcast transmissions
- Explain why IPv6 addressing is needed.
- Explain the difference between an IPv6 Global Unicast Address and an IPv6 Link-local Address.
- Describe the IPv6 Global Unicast Address structure and the role each part plays.
- Explain how host-to-host connectivity is tested and how the PING command works.
- Explain the difference in PING and TRACERT. When would you use one over the other?

Step 8 – Presentation

- Present your completed project in a presentation format to your workgroup and demonstrate full connectivity.